KRISHI SETU

in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in

COMPUTER SCIENCE AND BUSINESS SYSTEMS

Submitted by

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Under the guidance of

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Department of Computer Science and Business Systems

ARISE & SHINE

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(An Autonomous Institution)

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CERTIFICATE

Certified that this project report KRISHI SETU is the Bonafide work of Sudeep Kuralkar and Himanshu Sherje who carried out the project work under my supervision in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science and Business Systems.**

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DEPARTMENT OF COMPUTER SCIENCE & BUSINESS SYSTEMS

KRISHI SETU

DIGITAL MARKETPLACE FOR DIRECT FARMER–BUYER INTERACTION AND INTELLIGENT CROP DECISION SUPPORT

Mapping of Micro Project with Program Outcomes (POs) and Program Specific Outcomes (PSOs)

POs												PSOs		
01	02	03	04	05	06	07	08	09	10	11	12	01	02	03
3	**2**	**1**	**2**	**2**	**1**	**2**	**1**	**2**	**3**	**2**	**2**	**2**	**3**	**2**

Prof. Renuka Raut (Guide)
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CONTENTS

ABSTRACT

Agricultural markets in developing economies continue to face significant challenges such as lack of transparency, inefficient supply chains, and heavy dependence on intermediaries. These issues limit farmers' ability to receive fair pricing for their produce and restrict buyers from accessing reliable sources. The absence of real-time market information further compels farmers to rely on traditional cultivation practices, often leading to imbalance between supply and demand.

Chapter 1: Introduces the fundamental problems in the agricultural sector and highlights the need for a digital solution that enables direct interaction between farmers and buyers. It also discusses the limitations of existing systems and establishes the motivation behind the proposed work.

Chapter 2: Presents the proposed system, Krishi Setu, a location-aware digital marketplace designed to bridge the communication gap between farmers and buyers. The system allows farmers to list crops and buyers to search based on location, price, and requirements. It also defines the objectives and user requirements of the system.

Chapter 3: Focuses on the analysis and design of the system, including entity relationships, module specifications, data flow, and database structure. The system architecture ensures efficient data management and real-time interaction. This chapter also includes implementation results and user interface representations.

Chapter 4: Summarizes the overall work and discusses the outcomes of the proposed system. It highlights the advantages of using a digital marketplace in agriculture and identifies limitations and future enhancements.

The system integrates geolocation services, real-time communication, and a data-driven price prediction mechanism that considers factors such as demand, supply, location, and historical trends. Additionally, it provides crop recommendations to assist farmers in making informed decisions.
The proposed solution aims to create a transparent, efficient, and scalable agricultural ecosystem that enhances market accessibility and improves farmer profitability while supporting sustainable agricultural practices.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION OF PROJECT

Agriculture continues to be a major contributor to economic development, particularly in developing countries. Despite its importance, the agricultural sector is affected by several challenges that reduce efficiency and profitability. One of the most critical issues is the absence of a structured and transparent marketplace that allows farmers to directly interact with buyers. This limitation has led to a dependency on intermediaries, who often influence pricing and reduce the share of profit received by farmers.

Another significant challenge is the lack of access to real-time market information. Farmers frequently rely on traditional practices and past experiences when selecting crops, without considering current demand trends. This results in an imbalance between supply and demand, leading to either surplus production or shortages in the market.

In recent years, mobile and internet technologies have created opportunities to digitize agricultural systems. However, most existing applications are limited to providing advisory services such as weather forecasts and crop guidance. These solutions do not address the core issue of market connectivity and direct trade.

To overcome these challenges, the proposed system, Krishi Setu, is developed as a location-aware digital marketplace. The platform enables farmers to list their crops and connect directly with buyers. Buyers can search for products based on location, price, and requirements, ensuring efficient and targeted interactions. The system also incorporates features such as real-time communication, price prediction, and crop recommendation to support informed decision-making.

The primary aim of the system is to create a transparent and efficient agricultural ecosystem that reduces dependency on intermediaries, improves market accessibility, and enhances farmer profitability.

1.2 LITERATURE REVIEW

Various digital solutions have been developed to support agricultural practices. Many of these systems focus on providing advisory services, including weather updates, soil analysis, and crop recommendations. While these applications contribute to improving awareness among farmers, they do not address the problem of direct market access.

Some supply chain platforms have been introduced to improve the distribution of agricultural products. These platforms aim to streamline logistics and reduce inefficiencies in transportation. However, they often operate through centralized systems that act as intermediaries, limiting the ability of farmers to negotiate prices directly with buyers.

Research in predictive analytics has explored the use of data-driven techniques for crop selection and yield prediction. Although these approaches provide valuable insights, they are typically not integrated with real-time market demand, which reduces their effectiveness in practical scenarios.

A key limitation observed in existing systems is the lack of integration between marketplace functionality, communication, and predictive analytics. Most platforms address only a specific aspect of agriculture, resulting in fragmented solutions.

The proposed system, Krishi Setu, addresses these gaps by integrating multiple functionalities into a single platform. It combines direct marketplace access, real-time communication, and intelligent decision-support mechanisms, providing a comprehensive solution for modern agricultural challenges.

1.3 SCOPE OF WORK

The scope of the Krishi Setu system includes the design and development of a mobile-based application that facilitates direct interaction between farmers and buyers. The system allows farmers to create and manage crop listings, including details such as crop type, quantity, price, and location.
Buyers can search for crops using various filters, such as location, price range, and crop category. The system provides a user-friendly interface that ensures ease of use for both farmers and buyers, even in rural environments.

The platform also includes a real-time communication module that enables negotiation between users. Additionally, a price prediction mechanism is incorporated to provide estimated market prices based on multiple factors. A crop recommendation feature is also included to assist farmers in making informed cultivation decisions.

The system is designed to be scalable and can be extended in the future to include additional features such as payment integration, logistics management, and advanced analytics.

1.4 OPERATING ENVIRONMENT – HARDWARE AND SOFTWARE (TECHNOLOGY USED)

The Krishi Setu system is developed using modern technologies to ensure performance, scalability, and usability.

Hardware Requirements:
The system requires an Android-based smartphone with a minimum of 4GB RAM and internet connectivity. The application is optimized to run efficiently on mid-range devices to ensure accessibility in rural areas.

Software Requirements:
The mobile application is developed using Android Studio with **React Native** and at core level to make efficient Java or Kotlin as the programming language. Node Js and Express Js is used as the backend platform for database management, authentication, and real-time data synchronization.

Map Libre API is integrated to provide location-based services, enabling users to search for crops based on geographical proximity. REST APIs are used for communication between the frontend and backend systems.

The system architecture is designed to support real-time updates and ensure reliable performance under varying network conditions.

CHAPTER 2

PROPOSED SYSTEM

2.1 OBJECTIVES OF PROPOSED SYSTEM

The proposed system, Krishi Setu, is designed to address the limitations of traditional agricultural markets by introducing a digital platform that enables direct interaction between farmers and buyers. The system focuses on improving transparency, efficiency, and accessibility in agricultural trade. The primary objective of the system is to eliminate the dependency on intermediaries by allowing farmers to directly connect with buyers. This ensures fair pricing and improves the income potential for farmers. Another key objective is to provide a transparent marketplace where users can view real-time information about available crops, pricing, and location. This helps both farmers and buyers make informed decisions.
The system also aims to assist farmers in selecting profitable crops by analyzing market demand and providing recommendations. This reduces the risk associated with traditional farming practices and promotes data-driven agriculture.

Additionally, the system is designed to enhance accessibility by providing a user-friendly mobile application that can be used in rural areas. The platform ensures that even users with limited technical knowledge can easily navigate and use the system.

The objectives of the proposed system can be summarized as follows:

- To enable direct communication between farmers and buyers
- To improve price transparency and reduce exploitation
- To provide location-based crop discovery
- To support intelligent decision-making through predictive insights
- To create a scalable and efficient digital agricultural marketplace

2.2 USER REQUIREMENT SPECIFICATION

The Krishi Setu system is designed to cater to three primary types of users: Farmers, Buyers, and Administrators. Each user category has specific requirements that are addressed by the system.

Farmer Requirements:
Farmers require a simple and efficient interface to manage their crop listings and interact with buyers. The system allows farmers to register and create profiles with relevant details such as location and contact information.
Farmers can add crop listings by providing information such as crop name, quantity, expected price, and availability. They can update or remove listings as needed.
The system also enables farmers to receive inquiries from buyers and communicate through a real-time chat module. Additionally, farmers receive price suggestions and crop recommendations based on market data, helping them make better decisions.

Buyer Requirements:

Buyers require an efficient mechanism to search and discover crops based on their specific needs. The system allows buyers to register and access a dashboard where they can search for crops using filters such as location, price range, and crop category.

Buyers can view detailed information about each listing, including farmer details and crop availability. The platform enables buyers to directly communicate with farmers, negotiate prices, and finalize transactions.
The location-based search feature ensures that buyers can find nearby suppliers, reduce transportation costs and improve efficiency.

Administrator Requirements:
The administrator is responsible for managing the overall system and ensuring smooth operation. The system provides an admin panel that allows administrators to monitor user activities, manage listings, and handle complaints or disputes.
The admin can approve or remove users and listings, ensuring that the platform maintains quality and reliability. The administrator also has access to analytical data that provides insights into market trends, user activity, and system performance.
Overall, the user requirements are designed to ensure that the system is functional, user-friendly, and capable of supporting efficient agricultural trade.

CHAPTER 3

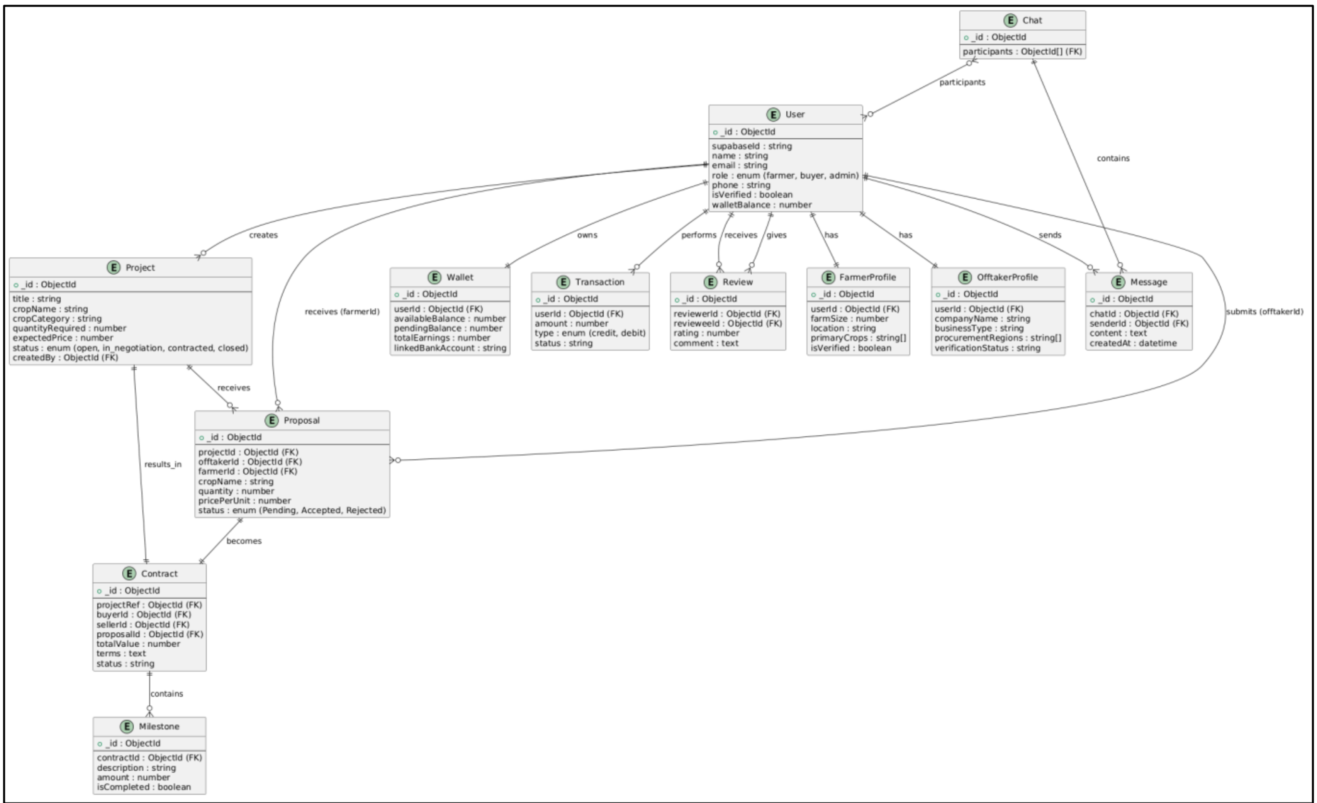
ANALYSIS AND DESIGN

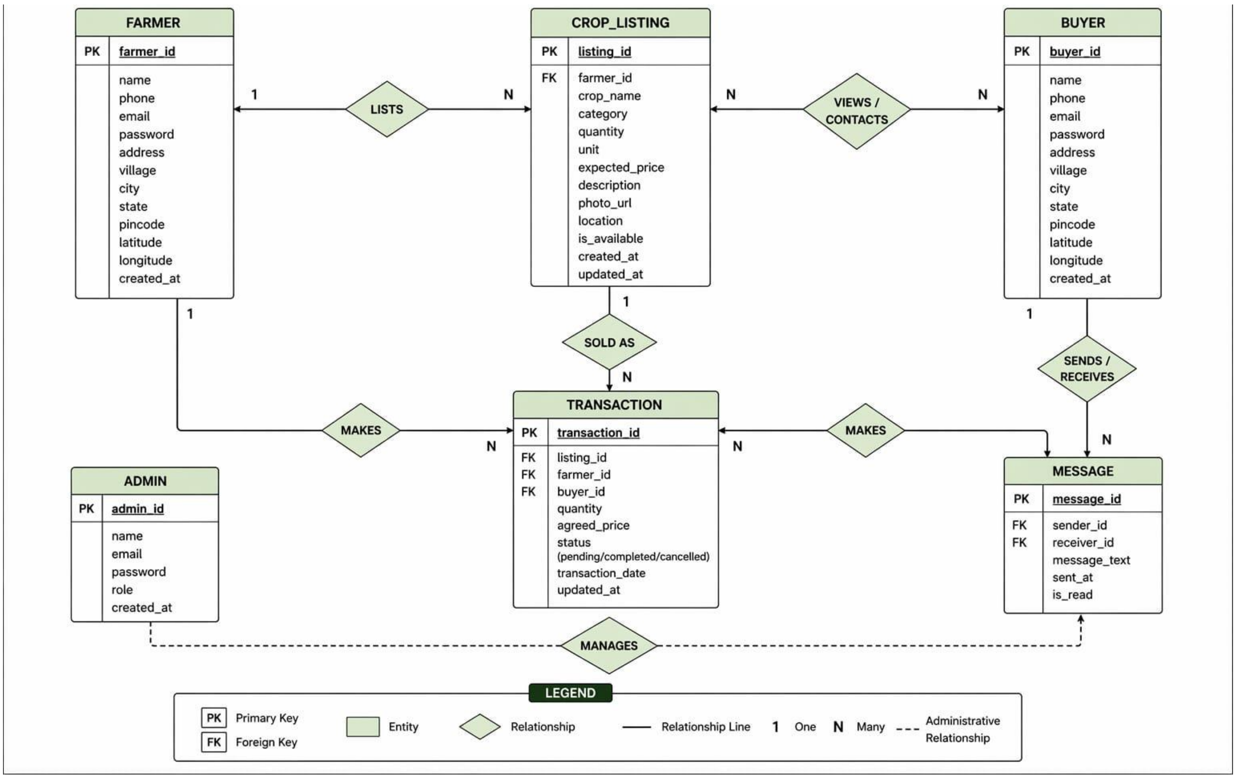
3.1 ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship (ER) diagram represents the structure of the database and the relationships between different entities in the system. The primary entities identified in the Krishi Setu system include Farmer, Buyer, Crop Listing, Transaction, and Admin.

The Farmer entity stores information related to farmers such as name, contact details, and location. The Buyer entity stores similar details for buyers. The Crop Listing entity contains information about the crops uploaded by farmers, including crop name, quantity, price, and availability.

The Transaction entity represents the interaction between farmers and buyers when a crop is sold. The admin entity manages user activities and system operations. The relationships between these entities ensure proper data flow within the system. For example, a farmer can create multiple crop listings, while a buyer can interact with multiple listings.





3.2 MODULE SPECIFICATION

The system is divided into multiple modules to ensure efficient functionality and maintainability.

Authentication Module:

This module handles user registration and login functionality. It ensures secure access to the system by validating user credentials and maintaining session data.

Marketplace Module:

This module allows farmers to create, update, and manage crop listings. Buyers can search for crops using filters such as location, price, and category. The module ensures smooth interaction between farmers and buyers.

Communication Module:

The communication module enables real-time chat between farmers and buyers. This allows users to negotiate prices and discuss product details before finalizing transactions.

Location Module:

This module integrates geolocation services to provide location-based filtering. Buyers can search for crops available in nearby areas, improving efficiency and reducing transportation costs.

Analytics Module:

The analytics module processes data to generate price suggestions and crop recommendations. It uses multiple factors such as demand, supply, location, and historical trends to provide insights.

Admin Module:

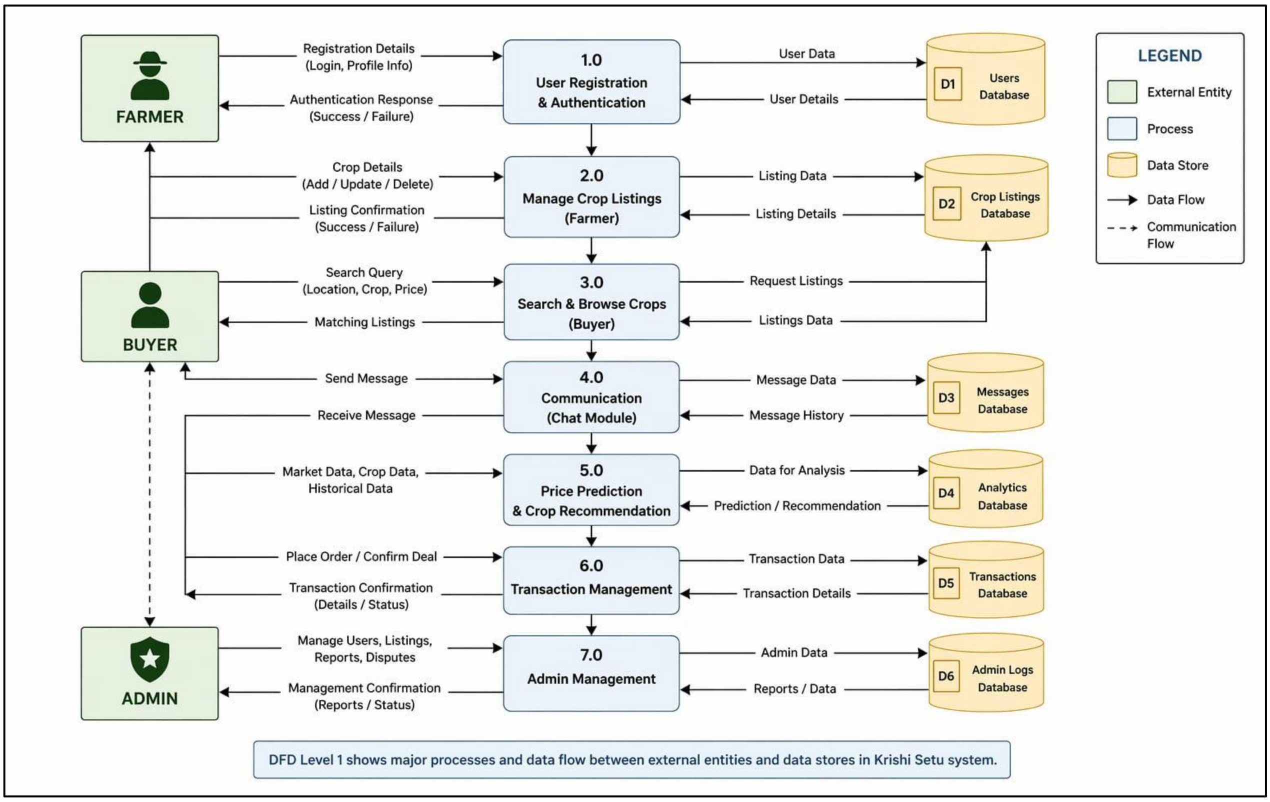
The admin module allows administrators to monitor system activities, manage users, and handle complaints. It ensures the reliability and security of the platform.

3.3 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) represents the flow of data within the system. It shows how data moves between users, processes, and the database. The process begins with user registration, where user details are stored in the database. Farmers upload crop information, which is also stored in the system. Buyers retrieve this data through search queries.

Communication between farmers and buyers occurs through the chat module. The system also processes data to generate price suggestions and recommendations.

The DFD helps in understanding the interaction between different components of the system and ensures efficient data management.



3.4 TABLE SPECIFICATION

The database is designed to store and manage system data efficiently. The main tables used in the system are as follows:

User Table:
This table stores user information including user ID, name, role (farmer or buyer), contact details, and location.

Crop Listing Table:
This table stores details of crops uploaded by farmers, including crop ID, farmer ID, crop name, quantity, price, and availability.
Transaction Table:

This table records transactions between farmers and buyers. It includes transaction ID, buyer ID, farmer ID, crop ID, and transaction status.

Chat Table:
This table stores messages exchanged between users. It includes message ID, sender ID, receiver ID, message content, and timestamp.

The database structure is designed to ensure data consistency, efficient retrieval, and scalability.

3.5 RESULTS

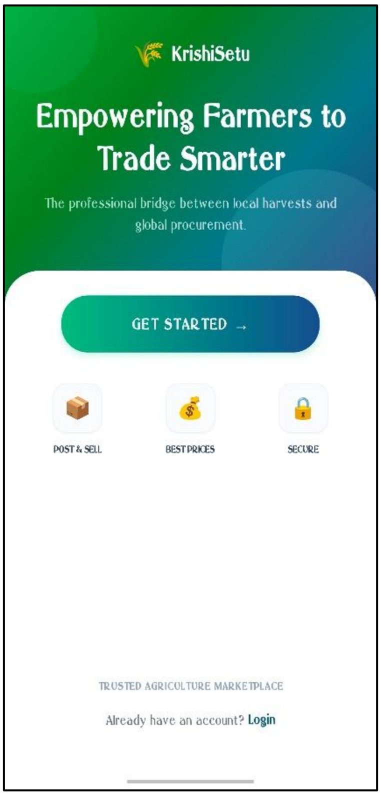
The implementation of the Krishi Setu system consists of multiple user interfaces designed to support different functionalities for farmers and buyers (off takers). Each interface is developed to ensure usability, efficiency, and real-time interaction within the system.

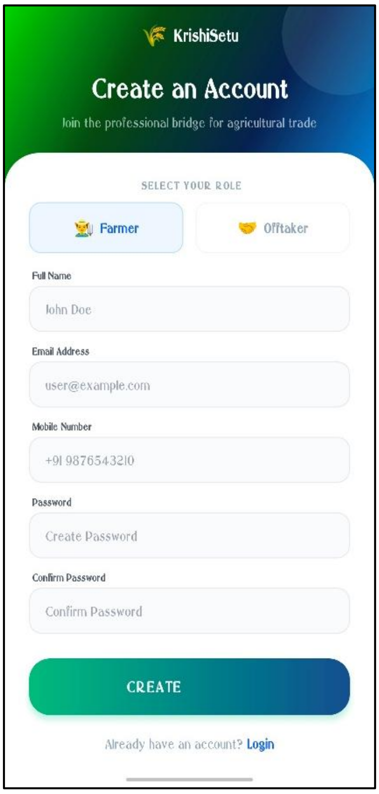
The following sections describe the major screens of the application along with their functionalities.

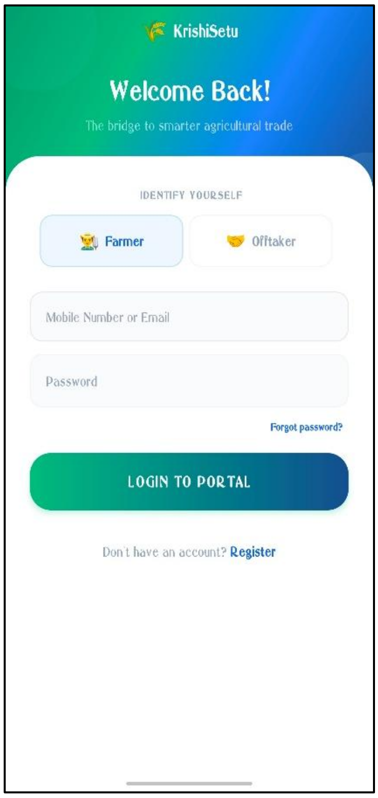
LOGIN AND REGISTRATION MODULE

The login and registration interface serves as the entry point to the system. It allows users to securely create accounts and access the platform based on their roles (Farmer or Buyer).

The registration process requires users to provide essential details such as name, contact number, location, and role selection. Authentication is handled securely using backend services, ensuring data







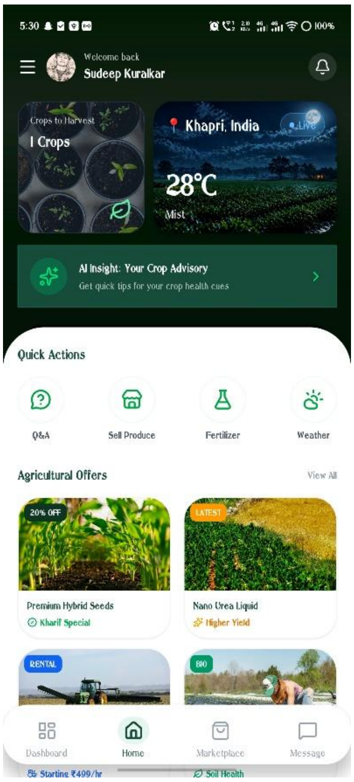
protection and controlled access. The login system validates user credentials and redirects users to their respective dashboards based on their role.

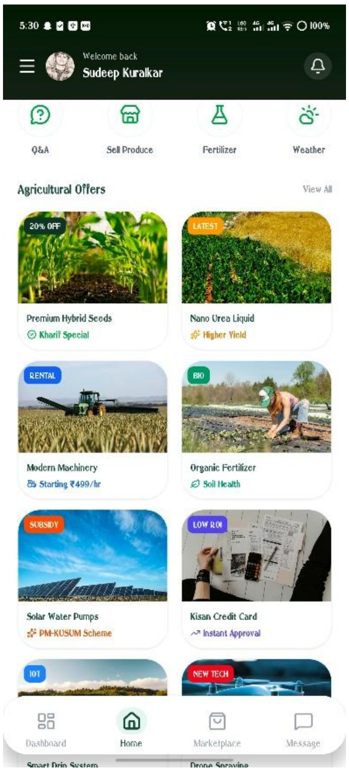
FARMER MODULE

The farmer module is designed to provide complete control over crop management and interaction with buyers. It includes several features that assist farmers in decision-making and marketplace participation.

Farmer Dashboard:

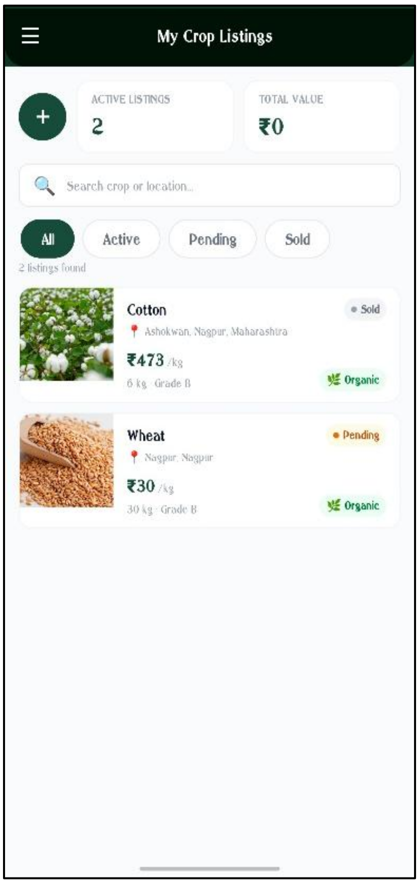
The farmer dashboard provides an overview of all activities, including active listings, buyer inquiries, and recent interactions. It acts as the central control panel for managing crop-related operations.

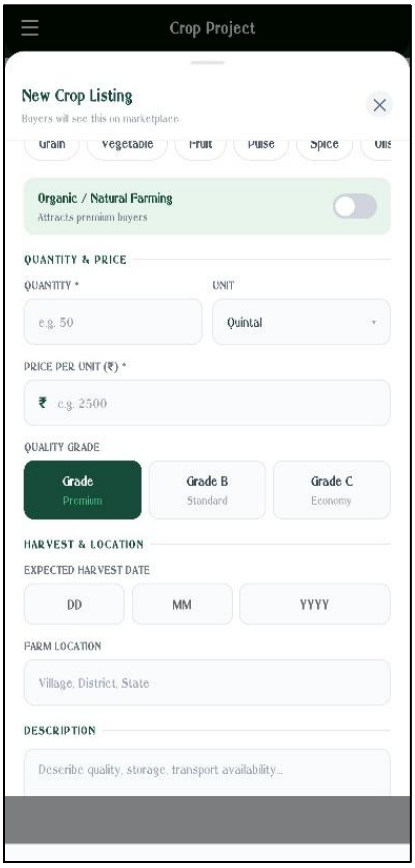


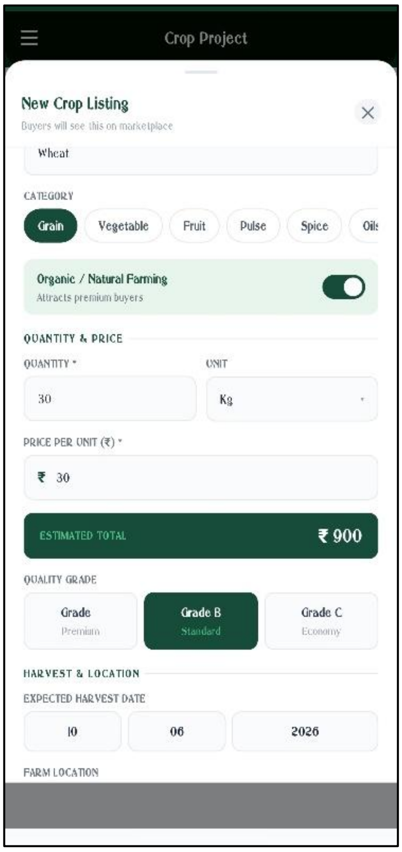


Crop Listing Page:

This page serves as a central interface for farmers to upload and manage their crop listings within the system. It allows farmers to enter detailed information such as crop name, quantity, expected price, and a descriptive overview, which helps buyers clearly understand the product being offered. The interface is designed to be user-friendly, ensuring that even users with limited technical knowledge can easily navigate and perform actions. In addition to creating new listings, farmers have the flexibility to update existing entries or remove them when crops are sold or no longer available. This dynamic control over listings ensures that the information remains accurate and relevant at all times. The system processes and stores all data efficiently in the backend, while real-time synchronization ensures that any changes made by farmers are instantly reflected across the platform. This not only improves data consistency but also enhances transparency and trust between farmers and buyers. By providing a seamless and responsive listing management experience, this page plays a crucial role in enabling effective digital marketplace interactions.





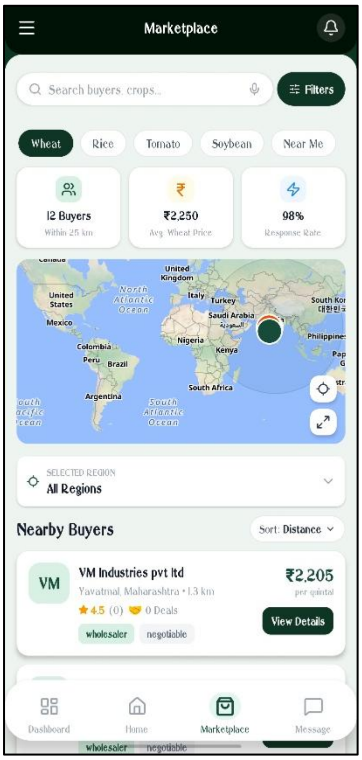


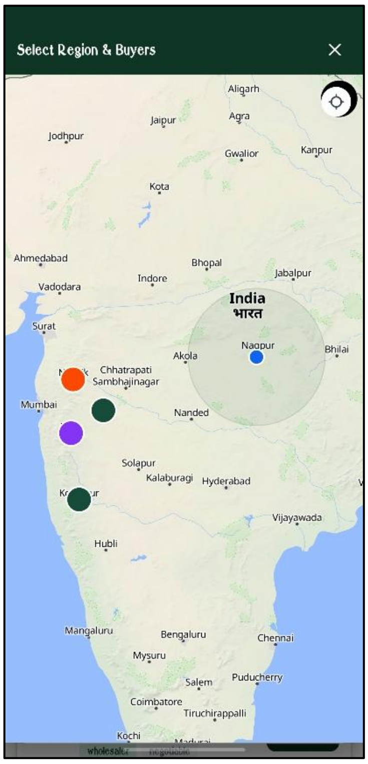
Weather Information Module:

This module provides real-time weather updates based on the farmer's current location, enabling them to make informed agricultural decisions. It displays essential weather parameters such as temperature, humidity, rainfall, and wind conditions, along with short-term forecasts. By accessing accurate and timely weather information, farmers can efficiently plan activities like sowing, irrigation, fertilization, and harvesting. Additionally, the module helps in minimizing risks caused by unexpected climatic changes by providing alerts for extreme weather conditions, ultimately improving crop productivity and resource management.

Location Services:

This module utilizes GPS-based location services to accurately capture and tag the geographical position of farmers and their crop listings. By enabling location-based filtering, it allows buyers to easily discover nearby farmers and available produce, ensuring faster and more efficient connections. This feature enhances visibility for local farmers, reduces transportation costs, and supports quicker transactions, ultimately contributing to a more efficient and localized agricultural marketplace.



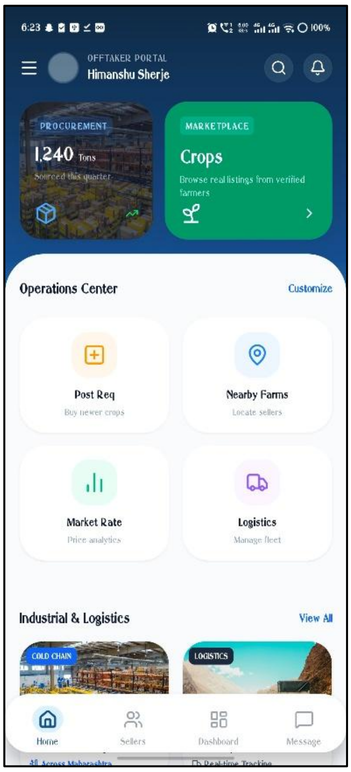


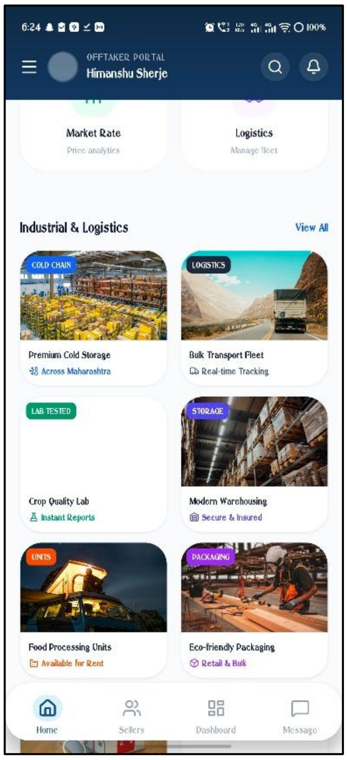
BUYER (OFFTAKER) MODULE:

The buyer module is designed to help users efficiently search for and procure agricultural products. It provides filtering, communication, and decision-support features.

Buyer Dashboard:

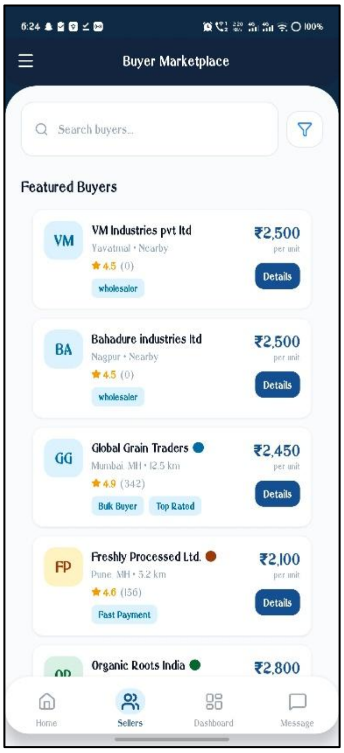
The buyer dashboard serves as a centralized interface that displays available crop listings, recent searches, and communication history in a structured and user-friendly manner. It allows buyers to quickly browse and filter crops based on their preferences, view detailed information about listings, and reconnect with previously contacted farmers. Additionally, the dashboard provides quick access to all major functionalities such as search, messaging, and profile management, ensuring a smooth and efficient user experience. By organizing





essential features in one place, it helps buyers make faster decisions and enhances overall usability of the platform.

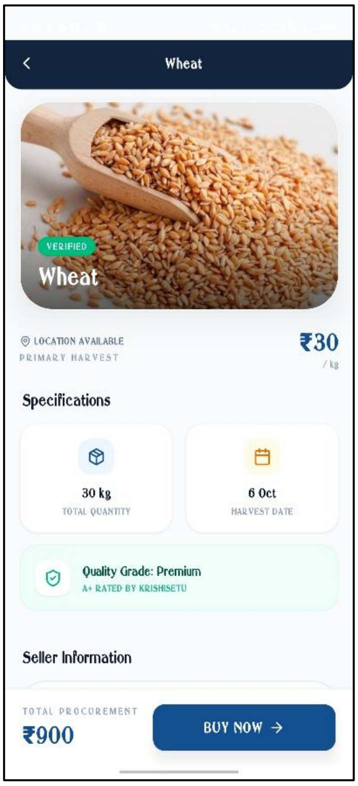
Search and Filter Page:

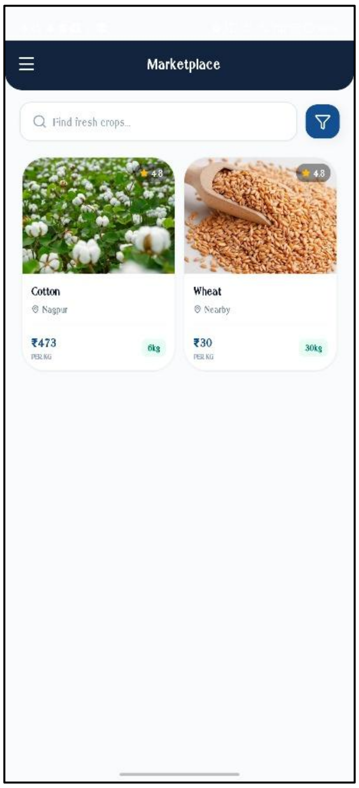


Buyers can search for crops using multiple parameters such as location, crop type, price range, and availability, making the process highly flexible and user-friendly. The system is designed to handle detailed filtering, allowing buyers to narrow down their options based on specific needs, whether they are looking for a particular variety of crop, a preferred region, or a budget constraint. Additionally, real-time availability ensures that buyers are only viewing active and relevant listings, reducing time wasted on outdated information. This advanced filtering mechanism not only improves search efficiency but also enhances the overall user experience by helping buyers quickly discover suitable products that match their exact requirements, ultimately facilitating faster and more effective transactions between farmers and buyers.

Listing Details Page:

This page provides detailed information about a selected crop, including farmer details, pricing, quantity, and location. Buyers can use this information to make informed purchasing decisions.

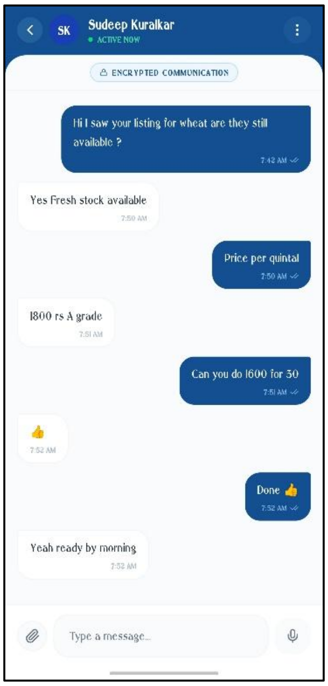


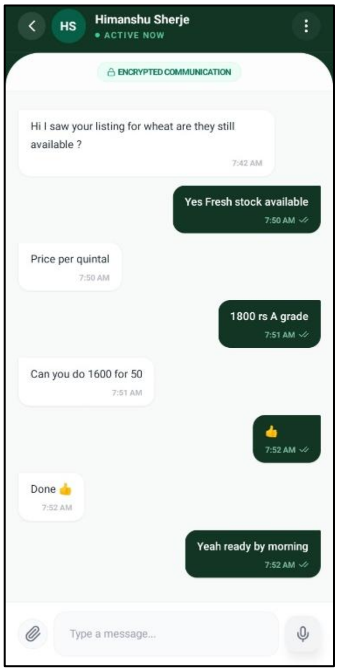


COMMUNICATION MODULE

Chat Interface:

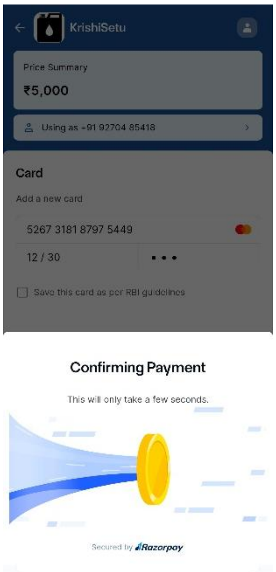
The chat module enables real-time communication between farmers and buyers. It allows users to negotiate prices, clarify details, and finalize transactions. The system ensures smooth message exchange and maintains chat history for future reference.

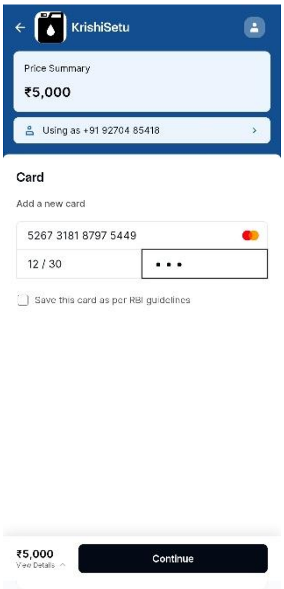


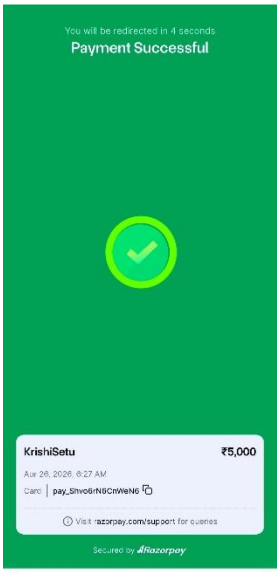


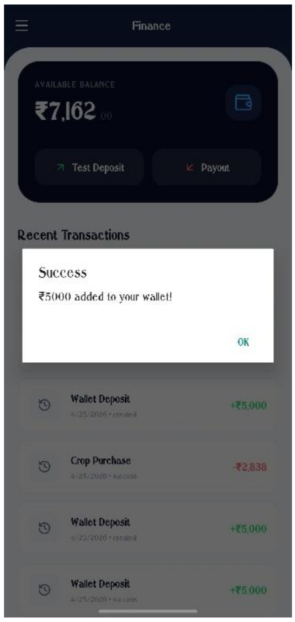
TRANSACTION INTERACTION:

If transaction confirmation or order flow is implemented, this section shows how buyers confirm deals and farmers receive confirmation.









Overall, the results demonstrate that the Krishi Setu system successfully integrates multiple functionalities into a unified platform. The application provides a seamless experience for both farmers and buyers, improving communication, decision-making, and market accessibility.

The integration of Razor pay in the Krishi Setu platform plays a crucial role in ensuring seamless, secure, and efficient financial transactions between buyers and farmers. By leveraging Razor pay's robust payment infrastructure, the application enables users to complete payments quickly using multiple options such as UPI, debit/credit cards, net banking, and digital wallets. This flexibility is particularly important in an agricultural marketplace where users may have varying levels of digital payment access and preferences. Additionally, Razor pay's real-time payment processing ensures instant confirmation, reducing delays and building trust between both parties during transactions.

From an efficiency standpoint, Razor pay significantly simplifies the payment workflow within Krishi Setu by automating key processes such as payment verification, transaction tracking, and failure handling. Its high-level security standards, including encryption and compliance with industry regulations, protect sensitive financial data and minimize fraud risks. Furthermore, Razor pay's APIs allow smooth backend integration, enabling developers to maintain a reliable and scalable system even as user traffic grows. This results in a faster, more dependable transaction experience, ultimately enhancing user satisfaction and encouraging more frequent and confident use of the platform.

CHAPTER 4

CONCLUSION

4.1 SUMMARY OF WORK

The Krishi Setu system has been developed to address the major challenges faced in agricultural markets, particularly the lack of transparency and dependency on intermediaries. The project focuses on creating a digital platform that enables direct interaction between farmers and buyers, thereby improving efficiency and accessibility in agricultural trade.

The system successfully integrates multiple functionalities, including user authentication, crop listing, buyer search, real-time communication, and location-based filtering. These features work together to create a seamless and user-friendly experience for both farmers and buyers.

The inclusion of a price prediction mechanism and crop recommendation feature enhances the decision-making capabilities of farmers. By providing insights based on demand, supply, and historical data, the system encourages data-driven agricultural practices.

The implementation of the system demonstrates that digital platforms can significantly improve market transparency and reduce inefficiencies in traditional agricultural systems. The results indicate that the proposed system has the potential to increase farmer profitability and improve buyer accessibility.

Overall, the project achieves its objectives by providing a scalable and efficient solution for agricultural market challenges.

4.2 FUTURE SCOPE

The Krishi Setu system can be further enhanced by incorporating additional features and technologies to improve its functionality and reach.

One potential improvement is the integration of online payment systems, which would enable secure and seamless transactions between farmers and buyers. This would reduce dependency on cash-based transactions and improve reliability.

The system can also be extended by incorporating machine learning algorithms to improve the accuracy of price prediction and crop recommendation. Advanced models can analyze larger datasets to provide more precise insights.

Another important enhancement is the integration of Internet of Things (IoT) devices for real-time monitoring of agricultural conditions such as soil quality, temperature, and humidity. This would provide farmers with additional data to support decision-making.

The development of an offline mode can improve accessibility in areas with limited internet connectivity. Additionally, multilingual support can be introduced to make the system more user-friendly for diverse user groups.

The platform can also be expanded to include logistics and transportation management, enabling complete end-to-end agricultural supply chain solutions.

4.3 LIMITATIONS

Despite its advantages, the Krishi Setu system has certain limitations that need to be addressed in future development.

The system relies on internet connectivity, which may limit its usability in remote areas with poor network coverage. This can affect real-time communication and data access.

The accuracy of the price prediction model depends on the availability and quality of data. Limited or outdated data may reduce the effectiveness of predictions.

User adoption is another challenge, as farmers may require training and awareness to effectively use the system. Digital literacy plays a crucial role in the successful implementation of such platforms.

Additionally, the system currently focuses on direct interaction and does not include integrated payment or logistics features, which may be necessary for a complete marketplace solution.

Addressing these limitations in future versions of the system will further enhance its effectiveness and usability.

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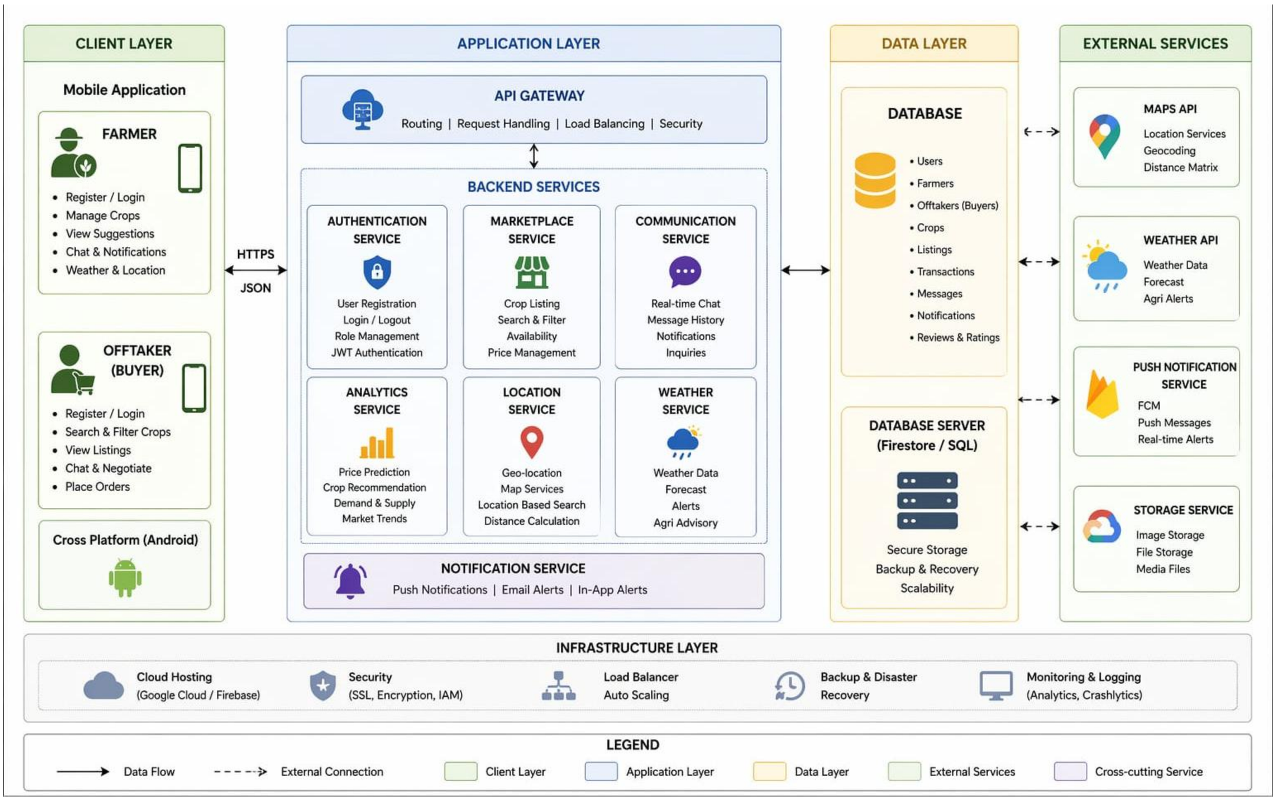
APPENDICES

APPENDIX I
SYSTEM ARCHITECTURE DIAGRAM

This appendix presents the overall architecture of the Krishi Setu system. The architecture follows a client–server model where the mobile application interacts with backend services for data processing, business logic execution, and secure data storage. The system is modular in design, consisting of key components such as authentication, marketplace, communication, analytics, weather information, and location services, each handling specific functionalities.

The frontend, developed using React Native, provides an intuitive and user-friendly interface for both farmers and buyers, while the backend manages APIs, database operations, and real-time data exchange. Communication between the client and server is handled through secure RESTful APIs, ensuring efficient and reliable data flow. Additionally, the architecture is designed to be scalable and flexible, allowing easy integration of new features and services in the future, thereby supporting the growing needs of the agricultural ecosystem.

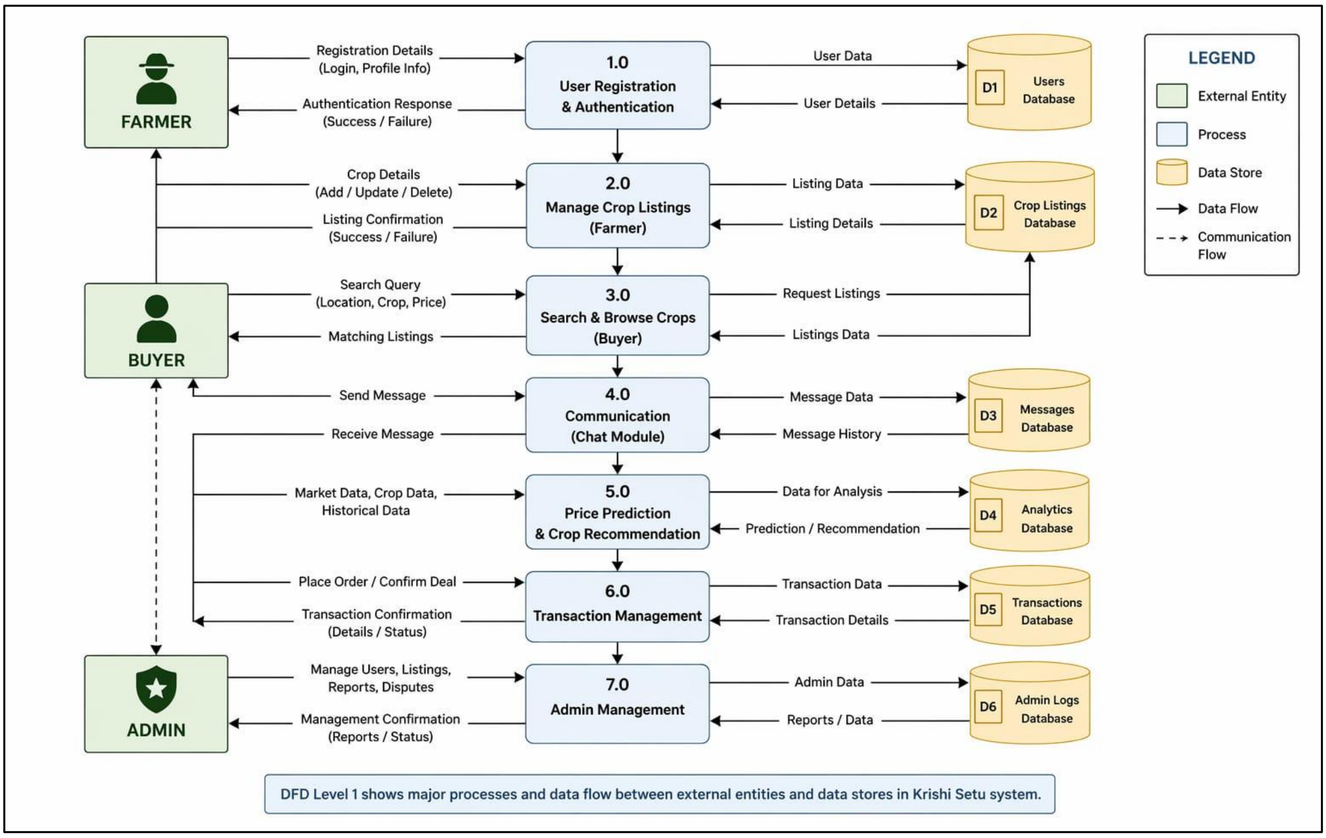


APPENDIX II

DATA FLOW DIAGRAM

This appendix illustrates the flow of data within the Krishi Setu system, providing a clear representation of how information moves between users, system processes, and the database. It demonstrates how user inputs are captured through the mobile application and processed by the backend before being securely stored or retrieved from the database. The diagram highlights key processes such as user authentication, crop listing, search functionality, and real-time communication between farmers and buyers.

Furthermore, it explains the interaction between different modules and how data is validated, processed, and delivered to ensure accuracy and efficiency. This structured data flow helps in understanding the system's functionality, improves transparency in operations, and ensures smooth coordination between various components of the platform.

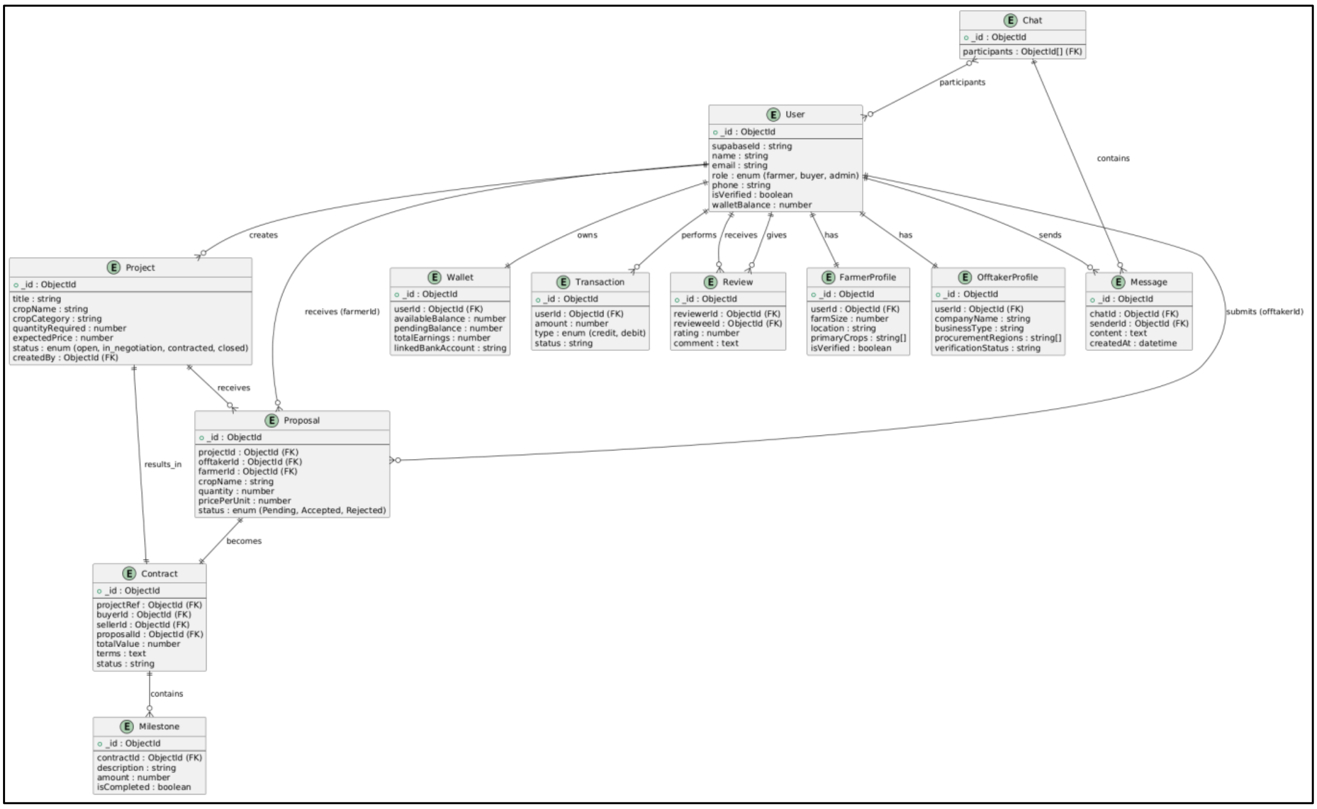


This diagram represents the **Level 1 Data Flow Diagram (DFD)** of the Krishi Setu system, illustrating how data moves between external entities, processes, and data stores. The primary external entities are **Farmer**, **Buyer**, and **Admin**, each interacting with different system processes. The flow begins with **User Registration & Authentication (1.0)**, where users provide login and profile details that are stored in the **Users Database (D1)**. Farmers then use **Manage Crop Listings (2.0)** to add, update, or delete crop details, which are stored in the **Crop Listings Database (D2)**. Buyers interact with the system through **Search & Browse Crops (3.0)**, sending queries based on location, crop type, or price, and receiving matching listings from the database. Communication between buyers and farmers is handled by the **Communication Module (4.0)**, where

messages are exchanged and stored in the **Messages Database (D3)**.
The system also includes advanced features like **Price Prediction & Crop Recommendation (5.0)**, which uses market data, historical trends, and crop data to generate insights stored in the **Analytics Database (D4)**. Once a buyer decides to proceed, the **Transaction Management (6.0)** module handles order placement, deal confirmation, and transaction tracking, with all financial records maintained in the **Transactions Database (D5)**. Finally, the **Admin Management (7.0)** module allows administrators to oversee the platform by managing users, listings, reports, and disputes, with logs stored in the **Admin Logs Database (D6)**. Overall, the DFD clearly shows a structured and efficient flow of data, ensuring smooth interaction between users and reliable backend processing across all major functionalities of the Krishi Setu platform.

APPENDIX III
ENTITY RELATIONSHIP DIAGRAM

This appendix shows the database structure of the system, including entities such as Farmer, Buyer, Crop Listing, Transaction, and Admin. The relationships between these entities ensure proper data management.



This ERD represents the core database structure of the Krishi Setu system, showing how different entities interact to support the platform's functionality. At the centre of the system is the **User** entity, which acts as a generalized model for all types of users, including farmers, buyers, and admins. Each user has attributes such as name, email, phone number, role, and verification status. Based on their role, users are further extended into specialized profiles like **Farmer Profile** and **Offtake Profile**. The Farmer Profile stores details such as farm size, location, and primary crops, while the Offtake Profile (buyer) includes company-related information, procurement regions, and verification status. This separation ensures role-specific data is managed efficiently without duplicating user information.

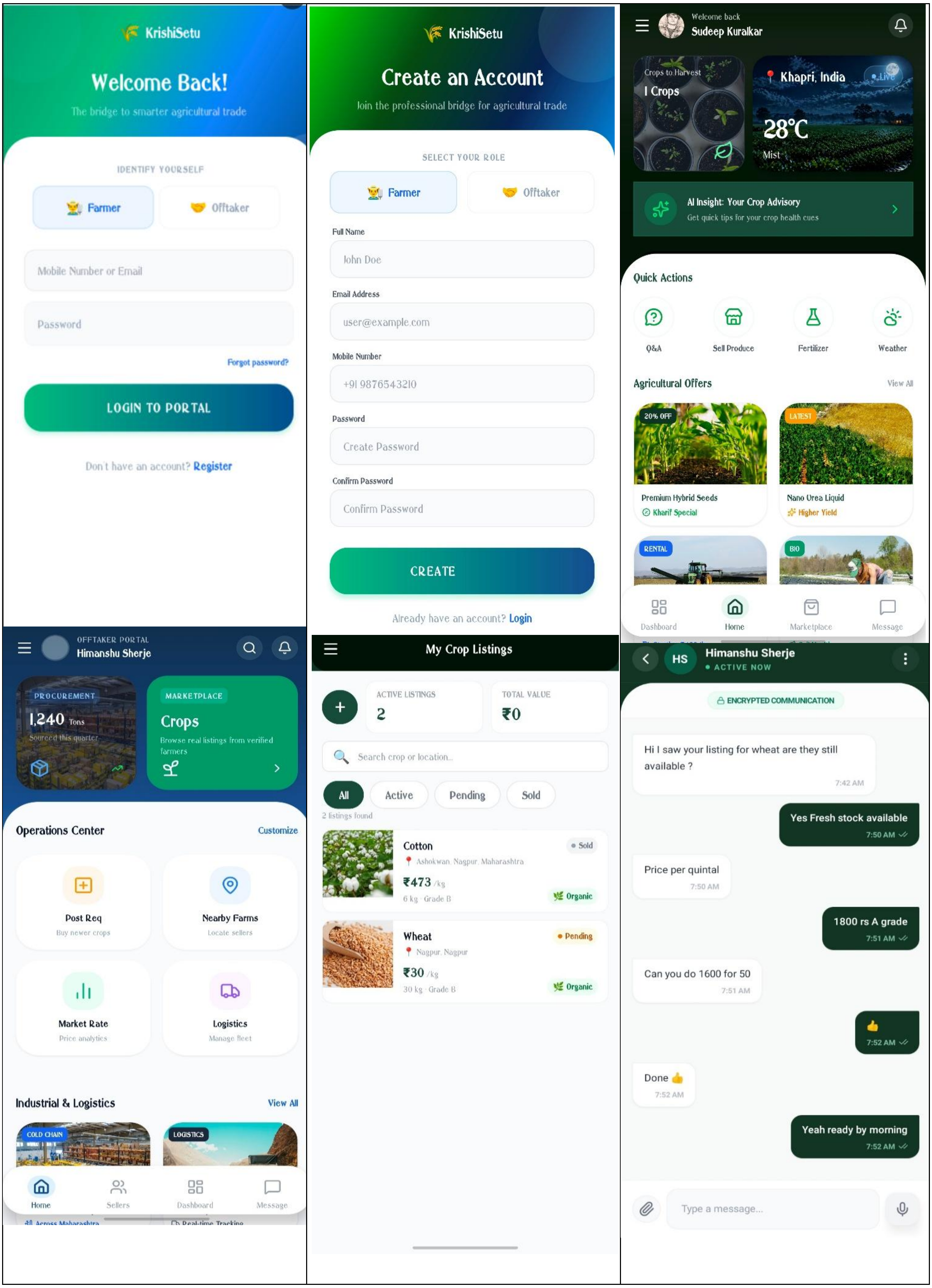
The **Project** entity represents crop listings created by farmers. It includes details such as crop name, quantity, expected price, and status (e.g., open, in negotiation, contracted, closed). Buyers interact with these listings by submitting **Proposals**, which contain offered price, quantity, and status (pending, accepted, rejected). Once a proposal is accepted, it evolves into a **Contract**, formalizing the agreement between buyer and farmer. Contracts may include multiple **Milestones**, allowing structured progress tracking for delivery or payment stages. This flow—Project → Proposal → Contract → Milestone—clearly models the lifecycle of a transaction in the system.

Financial interactions are handled through the **Wallet** and **Transaction** entities. Each user has a wallet that tracks available balance, pending balance, and linked bank account details. Transactions record all monetary activities, including credits and debits, along with their status. This ensures transparency and traceability of payments within the platform. Additionally, the **Review** entity allows users to rate and provide feedback on each other after transactions, helping build trust and credibility in the marketplace.

Communication between users is managed through the **Chat** and **Message** entities. A chat contains multiple participants (users), and each chat holds multiple messages. Messages include sender details, content, and timestamps, enabling real-time or asynchronous communication between farmers and buyers. This feature is crucial for negotiation and clarification before finalizing deals.

Overall, the relationships in this ERD are well-structured and normalized. A single user can own multiple projects, send multiple proposals, participate in multiple chats, and perform multiple transactions. The use of foreign keys ensures referential integrity across entities, while modular components like profiles, wallets, and messaging make the system scalable and maintainable. This design effectively supports the end-to-end workflow of an agricultural marketplace, from listing crops to completing secure transactions and maintaining communication.

APPENDIX IV (Application Screenshots)



APPENDIX V
PROJECT DOCUMENTS AND CERTIFICATES

Project Documents	Status	Plagiarism	Link
Project Report	Done	-	
Synopsis (Copyright)	Done	4%	Diary No: **LD-20068/2026-CO**
Android Application	Version 1 is ready	-	**https://sudeep042006.github.io/KrishiSetu/**
Research Paper	Done	4%	

This appendix presents a summary of the current status of all major project components related to the KrishiSetu application. The project report and research paper have been successfully completed with minimal plagiarism levels of 4%, ensuring originality and academic integrity. The synopsis has also been finalized and documented under Diary Number LD-20068/2026-CO, maintaining compliance with institutional requirements. Additionally, the Android application has reached Version 1 and is ready for deployment, with its functional prototype accessible via the provided GitHub Pages link. Overall, the project demonstrates steady progress across documentation, development, and research aspects.

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